UNIVERSITÄT PASSAU
Fakultät für Informatik und Mathematik

Search-Based Software Engineering

Organisational
Winter 2026/2027

Patric Feldmeier



Patric

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Timetable

- Lectures:
  - Tuesday 08:00–10:00
  - (IM) HS 11
  - Prof. Dr. Gordon Fraser
- Exercises:
  - Wednesday 08:00 - 10:00
  - (IM) SR 07
  - Patric Feldmeier

Grading

- Four assignments requiring you to implement search-based algorithms.
- Grade is based on portfolio of assignment solutions and exam
- Written or oral exam - depends on number of participants
- Passing exam is requirement for passing the course

Course Registration

- Registration window: from now on **until 2026–11–03**
- Registration is *binding* (there's no going back)
- $\geq$ 50% points over all four assignments & passing of oral/written exam required for passing course

Testing

Or: “Why Testing is Important”

- Quality of submission: Your responsibility (i.e., write test cases!)
- Part of the grade for an assignment will be determined by testing (e.g. coverage, mutation score)

The Assignments

A1: Test Case Prioritisation

- Algorithm: Simulated Annealing & Evolutionary Algorithm
- Release: 2026–11–10
- Introduction: 2026-11-11
- Due: 2026-11-30

A2: Test Suite Minimisation

- Algorithm: NSGA-II
- Release: 2026–12–01
- Introduction: 2026–12-02
- Due: 2026–12-21

A3: Neuroevolution

- Algorithm: Testing Games with Neatest
- Release: 2026–12–22
- Introduction: 2026–12–22 (in the Lecture)
- Due: 🎄2027–01–18🎄

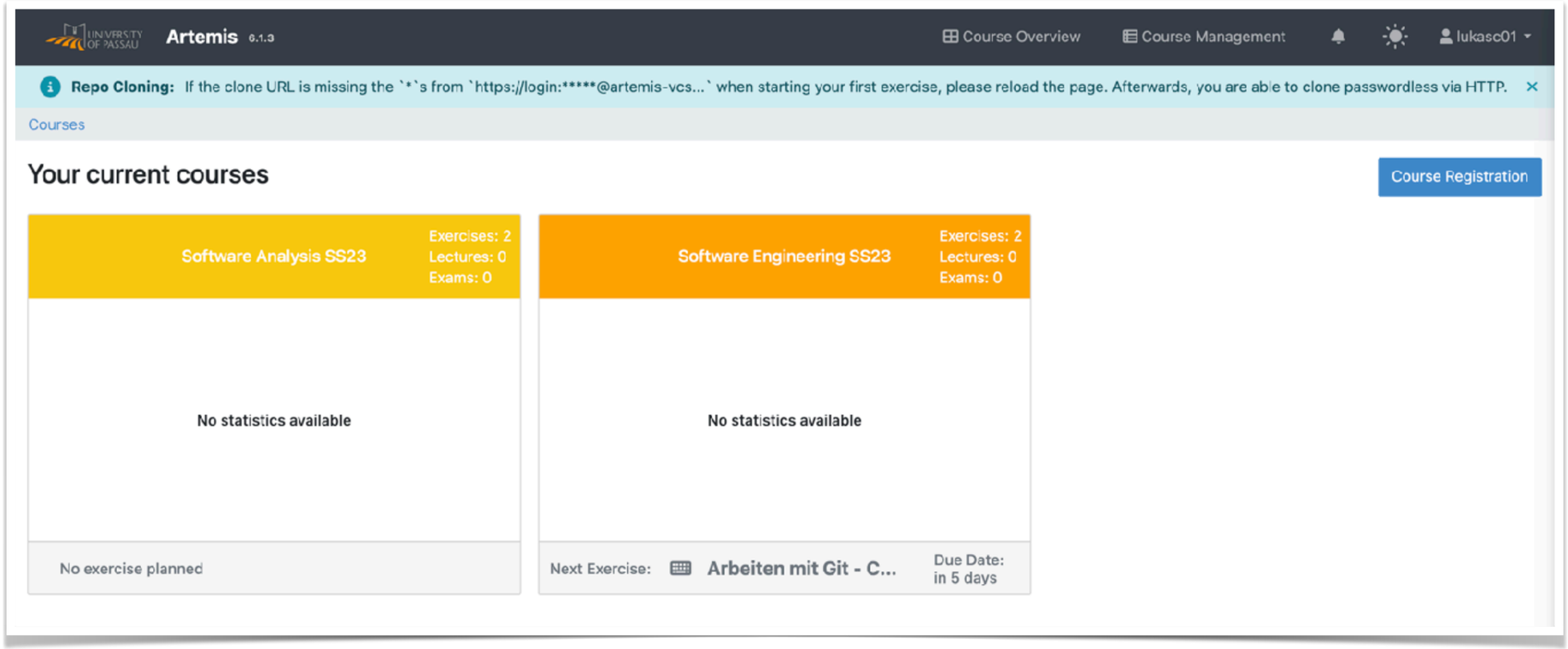
A4: Genetic Invariant Generation

- Algorithm: Evolutionary Search for Generating Invariants
- Release: 2027–01–19
- Introduction: 2027–01–20
- Due: 2027–02–15

Artemis

- We use Artemis as a submission platform
- https://artemis.fim.uni-passau.de
- Log in with ZIM credentials
- Join “Search-Based Software Engineering WS27/26” course
- HEAD of *main* branch at deadline:
  taken for our evaluation and grading
- Commit early and often!

Artemis…



- Artemis builds your submissions (like CI)
- Runs your tests
- Runs our (functional) tests
- Reports you the results

How does an Assignment look like?

- We provide code scaffold (Maven project)
- Task description on Artemis
- You implement functionality & unit tests
- Pushing commits: Artemis will compile code, run your tests, run our tests
- Results reported back into Artemis
- We provide an example that is not graded

Necessary Skills

- Profound knowledge of Java
- Testing software (JUnit 5, Mockito, etc.)
- Version control (git)

Rules

- Achieve sufficient number of points to pass course
- Hard coded solutions: no points at all
- **Exam Registration until 2026–11–03**
- No group work, assignments are single-person
- Characteristical parts of two or more solutions are identical => plagiarism
  - You are responsible for proving to us that you did not plagiarize
  - LLMs tend to create similar solutions -> keep logs of LLMs if you are using them
- In case of plagiarism: all involved students fail the course immediately

Questions

- The Stud.IP Forum
- In the exercise class
- Via E-Mail for personal things

